**Diego González Sánchez**

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`https://dglez91.github.io/`

## Employment

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| 11/2025 – Present | **MSCA postdoctoral researcher**, Institut de Mathématiques de Jussieu-Paris Rive Gauche (IMJ-PRG) attached to the Centre National de la Recherche Scientifique (CNRS), France |
| 11/2020 – 10/2025 | **Research fellow**, Alfréd Rényi Institute of Mathematics, Hungary |
| 02/2024 – 05/2024 | **Professor**, Budapest Semesters in Mathematics, Hungary |

## Education

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| 10/2016 – 10/2020 | **Ph.D.**, Autonomous University of Madrid, Spain<br>Thesis: *Topics in additive combinatorics and higher order Fourier analysis*<br>Supervisor: Prof. Pablo Candela. Defended on 25/09/2020.<br>Final score: Outstanding (*sobresaliente*) cum laude. |
| 09/2015 – 07/2016 | **M.A.St.**, in Pure Mathematics (Part III), Cambridge University, UK<br>Final score: 73/100, Merit |
| 09/2009 – 06/2015 | **B.Sc.** in Mathematics, Autonomous University of Madrid, Spain.<br>Final score: 9.41/10 outstanding (*sobresaliente*)<br>**B.Sc.** in Computer Science, Autonomous University of Madrid, Spain.<br>Final score: 9.41/10 outstanding (*sobresaliente*) |

## Visiting periods (at least one month)

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| 05/2019 – 08/2019 | Visiting researcher, Alfred Renyi Institute of Mathematics, Hungary |
| 09/2013 – 07/2014 | Erasmus student, Technical University of Denmark, Denmark |

## Papers published or accepted for publication

[1] *Spectral algorithms in higher-order Fourier analysis*, (with P. Candela and B. Szegedy), accepted for publication in Forum of Mathematics, Sigma. Open access: arXiv:2501.12287.

[2] *On the inverse theorem for Gowers norms in abelian groups of bounded torsion*, (with P. Candela and B. Szegedy), accepted for publication at Discrete Analysis, Open access: arXiv:2311.13899.

[3] *Free nilspaces, double-coset nilspaces, and Gowers norms*, (with P. Candela and B. Szegedy). Math. Ann. 395, 23 (2026), Open access: arXiv:2305.11233.

[4] *On $\mathbb{F}_2^\omega$-affine-exchangeable probability measures*, (with P. Candela and B. Szegedy). Studia Mathematica **279**, 1–69, (2024). Open access: arXiv:2203.08915.

[5] *On measure-preserving $\mathbb{F}_p^\omega$ systems of order $k$*, (with P. Candela and B. Szegedy). Journal d'Analyse Mathématique (JAMA), (2025). Open access: arXiv:2311.13899.

[6] *On higher-order Fourier analysis in characteristic $p$*, (with P. Candela and B. Szegedy). Ergodic Theory and Dynamical Systems, **43**, 3971–4040, (2023). Open access: arXiv:2109.15281.

[7] *A refinement of Cauchy-Schwarz complexity*, (with P. Candela and B. Szegedy). European Journal of Combinatorics **106**, 103592 (2022). Open access: arXiv:2109.05965.

[8] *New estimates for exponential sums over multiplicative subgroups and intervals in prime fields*, (with D. Di Benedetto, M. Z. Garaev, V. C. Garcia, I. E. Shparlinski, and C. A. Trujillo). Journal of Number Theory **215**, 261–274 (2020). Open access: arXiv:2003.06165.

[9] *On sets with small sumset and $m$-sum-free sets in $\mathbb{Z}/p\mathbb{Z}$*, (with P. Candela and D. J. Grynkiewicz. Bulletin de la Société Mathématique de France **149** (1), 155–177 (2020). Open access: arXiv:1909.07967.

[10] *A note on the Bilinear Bogolyubov Theorem: Transverse and bilinear sets.* With P.-Y. Bienvenu and A. D. Martínez. Proceedings of the American Mathematical Society **148** (1), 23–31 (2020). Open access: arXiv:1811.09853.

[11] *On nilspace systems and their morphisms.*, (with P. Candela and B. Szegedy). Ergodic Theory and Dynamical Systems, **40**, (11), 3015 – 3029, (2020). Open access: arXiv:1807.11510.

[12] *A Plünnecke-Ruzsa inequality in compact abelian groups.*, (with P. Candela and A. de Roton). Revista Matemática Iberoamericana **35** (7), 2169–2186 (2019). Open access: arXiv:1712.07615.

## Conference papers

[13] *An algorithmic spectral approach to higher-order Fourier analysis*, (with P. Candela and B. Szegedy). Discrete Mathematics Days, Granada, Spain, 29 June - 1 July (2026). Open access: arXiv:2501.12287.

[14] *Higher-order Fourier analysis via spectral algorithms*, (with P. Candela and B. Szegedy). Proceedings of the 13th European Conference on Combinatorics, Graph Theory and Applications EUROCOMB'25 Budapest, August 25–29, (2025). Open access: arXiv:2501.12287.

[15] *A short proof of an inverse theorem in bounded torsion groups*, (with P. Candela and B. Szegedy). Discrete Mathematics Days, Alcalá de Henares, July 3-5, (2024). Open access: arXiv:2311.13899.

[16] *Optimal transport with $f$-divergence regularization and generalized Sinkhorn algorithm*, (with D. Terjék). International Conference on Artificial Intelligence and Statistics, 5135-5165 (2022). Open access: arXiv:2105.14337.

[17] *A Refinement of Cauchy-Schwarz Complexity, with Applications*, (with P. Candela and B. Szegedy). Extended Abstracts EUROCOMB'21, 293-298 (2021). Open access: arXiv:2109.05965.

[18] *A step towards the $3k-4$ conjecture in $\mathbb{Z}/p\mathbb{Z}$ and an application to $m$-sum-free sets*, (with P. Candela and D. J. Grynkiewicz). Acta Mathematica Universitatis Comenianae, **88** (3), 521–525, (2019). Open access: arXiv:1909.07967.

## Preprints

[19] *The Jamneshan-Tao conjecture for finite abelian groups of bounded rank*, (with P. Candela and B. Szegedy). Open access: arXiv:2601.08810.

[20] *An inverse theorem for all finite abelian groups via nilmanifolds*, (with P. Candela and B. Szegedy). Open access: arXiv:2512.17468.

[21] *MLPs at the EOC: Concentration of the NTK*, (with D. Terjék). Open access: arXiv:2501.14724.

[22] *MLPs at the EOC: Spectrum of the NTK*, (with D. Terjék). Open access: arXiv:2501.13225.

[23] *A framework for overparameterized learning*, (with D. Terjék). Open access: arXiv:2205.13507.

## Contributed talks at conferences and workshops

08/2025 *Higher-order Fourier analysis via spectral algorithms* at EuroComb 2025, Budapest, Hungary.

07/2025 *Inverse Gowers Theory via Ergodic Structure Theory* (together with Giorgio Baglioni and Nuno Carneiro) at 28th International Internet Seminar (ISEM28) "Ergodic structure theory and applications", Wuppertal, Germany.

07/2024 *A short proof of an inverse theorem in bounded torsion groups* at Discrete Mathematics Days 2024, Alcalá de Henares, Spain.

11/2023 *On measure-preserving $\mathbb{F}_p^\omega$ systems of order $k$* at the 11th workshop Operator Theoretic Aspects of Ergodic Theory, Wuppertal, Germany.

06/2023 *Free nilspaces, double-coset nilspaces, and Gowers norms* at Nilpotent structures in topological dynamics, ergodic theory and combinatorics, Będlewo Conference Center, Poland.

05/2022 *On affine-exchangeable measures and nilspaces* at the workshop Graphs, Groups, Stochastic Processes, Alfréd Rényi Institute of Mathematics, Budapest, Hungary.

09/2021 *A Refinement of Cauchy-Schwarz Complexity, with Applications*, at EuroComb 2021, online (Barcelona, Spain).

08/2019 *The $3k-4$ theorem in $\mathbb{Z}/p\mathbb{Z}$ and an application to $m$-sum-free sets* at EuroComb 2019, Univerzita Komenskeho, Bratislava, Slovakia.

12/2017 *La desigualdad de Plünnecke-Ruzsa en grupos abelianos compactos* at the LXVI Reunión de Comunicaciones Científicas of RSME and UMA, Buenos Aires, Argentina.

## Invited talks at conferences and workshops

01/2026 *Higher-order Fourier analysis via spectral algorithms* at Congreso Bienal de la Real Sociedad Matemática Española Alicante, Spain.

09/2025 *A spectral approach to higher-order Fourier analysis, Part I* at Approximate Algebraic Structures and Arithmetic Combinatorics, Mathematical Institute of the Serbian Academy of Sciences and Arts, Belgrade, Serbia.

## Invited talks at seminars

02/2025 *On higher-order Fourier analysis* at Department of Mathematics and Department of Quantitative Methods seminar, CUNEF universidad, Spain.

11/2024 *An algorithm for higher-order Fourier analysis* at Séminaire de Théorie des Nombres de Nancy-Metz, Institut Élie Cartan de Lorraine, France.

06/2024 *On measure-preserving $\mathbb{F}_p^\omega$-systems of order $k$.* at DAGGER seminar, University of Warwick, UK.

09/2022 *The Polyak-Lojasiewicz condition and overparameterized learning* (with D. Terjék) at Euler Seminar, Université catholique de Louvain, Louvain, Belgium.

03/2019 *Conjuntos grandes que se parecen a progresiones en grupos cíclicos de orden primo* (Large sets that resemble arithmetic progressions in cyclic groups of prime order), at Junior Seminar, Autonomous University of Madrid, Spain.

06/2018 *A Plünnecke-Ruzsa inequality in compact abelian groups* at UAM-ICMAT number theory seminar, ICMAT, Madrid, Spain.

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| 03/2017 | *El método polinomial en combinatoria aditiva* (the polynomial method in additive combinatorics) at the Junior Seminar, Autonomous University of Madrid, Spain. |

## Other events attended

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| 07/2023 | 200 Years of Trinity Combinatorics, Cambridge, UK. |
| 06/2022 | Graphs, Groups, Stochastic processes summer school, Budapest, Hungary. |
| 08/2019 | Number theory in the Americas. Casa Matemática Oaxaca, Oaxaca, Mexico. |
| 09/2018 | Arithmetic Ramsey Theory, University of Manchester, Manchester, UK. |
| 09/2017 | The music of Numbers. Conference in honor of Javier Cilleruelo. ICMAT, Madrid, Spain. |
| 05/2017 | Graduate course on Interactions of harmonic analysis, combinatorics and number theory. BGSMath, Barcelona, Spain. |

## Events organized

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| 06/2025 | New Trends in Arithmetic Combinatorics and related Fields, a BIRS-IMAG workshop. Coorganized with: P. Candela (lead organizer), A. de Roton, H. Helfgott, A. Sedunova, and O. Serra. Check our website: `https://birs-imag-ntac.github.io/`. |

## Projects in which I have participated and role

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| 11/2025 – 10/2027 | *Algorithmic higher-order Fourier analysis (HORIZON-MSCA-2024-PF-01, AlgHOF 101202161)*, European Comission. I am the IP, funding of a 2-year postdoc. |
| 09/2025 – 08/2028 | *Arithmetic, Combinatorics and Analysis (PID2024-156180NB-I00 )*, Agencia estatal de Investigación, MCIU, Spain. I am part of the working team. |
| 03/2022 – 12/2025 | *Artificial Intelligence National Laboratory (MILAB, RRF-2.3.1-21-2022-00004)*, NKFIH, Hungary. It partially funds the Research Fellow position at Alfréd Rényi institute of Mathematics. I work in the mathematical foundations of AI. |
| 09/2021 – 08/2024 | *Harmonic analysis, combinatorics and arithmetic* (PID2020-113350GB-I00) funded by MICINN, Spain. I am part of the working team. |
| 11/2020 – 11/2022 | *Momentum (Lendület) 30003*, NKFIH, Hungary. It partially funds the Research Fellow position at Alfréd Rényi institute of Mathematics. I work in HOF and in the mathematical foundations of AI. |
| 11/2020 – 03/2025 | *"Élvonal" KKP 133921 grant*, NKFIH, Hungary. It partially funds the Research Fellow position at Alfréd Rényi institute of Mathematics. I work in HOF and deep learning in this project. |
| 01/2018 – 12/2020 | *Arithmetic and harmonic analysis*, (MTM2017-83496-P) MINECO, Spain. I am part of the working team. |

## Research Fellowships/Prizes/Grants

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| 01/11/2025 | Marie Skłodowska-Curie postdoc (HORIZON-MSCA-2024-PF-01). |
| 15/07/2021 | Extraordinary award to the best theses presented at the Autonomous University of Madrid in the period 01/10/2019 to 30/09/2020. Awarded by the Autonomous University of Madrid. |

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| 10/02/2017 | La Caixa Severo Ochoa scholarship awarded by Fundación la Caixa (private company). Ph.D.'s funding. It included a monthly stipend, an amount for travels and other expenses, and all labour/tax costs. |
| 27/06/2015 | X Mutua Madrileña Posgraduate Scholarship awarded by Fundación Mutua Madrileña (private company). M.A.St.'s funding. |
| 18/06/2014 | JAE-intro scholarship for introduction to research awarded by CSIC, Ministry of Economy and Competitivity, Spain. |
| 03/12/2012 | Madrid Excellence Scholarship awarded by the Autonomous Community of Madrid, Spain. |
| 14/09/2011 | Madrid Excellence Scholarship awarded by the Autonomous Community of Madrid, Spain. |
| 26/01/2010 | Madrid Excellence Scholarship awarded by the Autonomous Community of Madrid, Spain. |

## Accreditations

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| 12/01/2022 | *Profesor ayudante doctor* (assistant professor) granted by the Ministry of Universities of Spain. |

## Graduate teaching

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| 08/2021 | Instructor at AGRA IV (Arithmetic, Groups and Analysis), online (Trieste, Italy). |

## Undergraduate teaching

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| 02/2024 – 05/2024 | Professor (jointly with P. Zsámboki) of *Introduction to Deep Learning* at the Budapest Semesters in Mathematics (BSM), Spring Semester 2024 (28 hours taught). Note: This was the first time this course was taught at BSM. Thus, Zsámboki and I had to prepare it from scratch. |
| 09/2019 – 11/2019 | Assistant professor of *Mathematics I* for the B.Sc. in Chemical engineering at the Autonomous University of Madrid (6 hours taught). |
| 09/2018 – 01/2019 | Assistant professor of *Mathematics* for the B.Sc. in Environmental Sciences at the Autonomous University of Madrid (15 hours taught). |
| 09/2017 – 01/2018 | Assistant professor of *Algebra I* for the B.Sc. in Physics at the Autonomous University of Madrid (15 hours taught). |

## Service

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| Referee for | TEMat, *Analysis and PDE*, *AISTATS 2022*, *AISTATS 2024* and *The Ramanujan Journal*. |
| Reviewer for | *Zentralblatt MATH (zbMath)*. |

## Other outreach activities

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| 22/04/2024 | *Recent trends III: Subsets of the integers without three term arithmetic progressions*, an outreach article for the Discrete and Algorithmic Mathematics (DAM) network. Available at: `https://dam-network.github.io/`. |

## Languages

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| Human: | Spanish (mother tongue)<br>English (fluent, used in professional contexts)<br>French (intermediate). |
| Programming: | Python (including Pytorch), C, Java, SQL, Matlab, and R. |